BAYES FILTER AND KALMAN FILTER FOR STATE ESTIMATION IN AUTONOMOUS VEHICLES

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Abstract—An autonomous vehicle (AV) cannot directly observe its exact position, velocity, orientation, or other dynamic states. Instead, it receives imperfect measurements from sensors and predicts motion using models that are themselves uncertain. Bayesian filtering provides a principled framework for combining these sources of information over time. This report introduces the recursive Bayes filter, defines its prediction and correction stages, and explains the conditions under which it becomes the linear Kalman filter. The Kalman filter equations are developed for a linear state-space model, with emphasis on covariance propagation, innovation, Kalman gain, and measurement correction. The report also discusses the consequences of nonlinearity and non-Gaussian uncertainty, and places filtering within an autonomous-driving perception and localisation pipeline. A practical sensor-fusion example involving inertial measurement, wheel odometry, and GNSS is used to show how filtering supports a stable vehicle-state estimate.

Keywords-Bayesian estimation, Bayes filter, Kalman filter, state estimation, sensor fusion, autonomous vehicles, localisation.

I. INTRODUCTION

State estimation is the process of inferring the internal state of a dynamic system from indirect, noisy observations. For a car, the state may include planar position, longitudinal and lateral velocity, heading angle, yaw rate, acceleration, and sensor biases. The vehicle does not have access to a perfectly accurate measurement of these quantities. A GNSS receiver may be affected by multipath and signal blockage; an inertial measurement unit (IMU) accumulates bias and drift; wheel encoders are affected by slip; and a camera or LiDAR may produce uncertain object or landmark measurements.

A motion model predicts how the vehicle state changes between observations. A sensor model describes how the state produces measurements. Neither is exact. A filter combines the prediction with new observations while explicitly representing uncertainty. The Bayes filter is the general probabilistic formulation. The Kalman filter is a computationally efficient, closed-form solution for the special case of linear dynamics, linear measurements, Gaussian noise, and a Gaussian initial state distribution [1], [2].

The distinction is important in autonomous driving (AD): a filter is not simply an averaging operation. It weights predictions and measurements according to their uncertainty, maintains a probability distribution over possible states, and updates that distribution recursively as data arrive.

II. THE BAYES FILTER

A. State, belief and conditional probability

Let x_t denote the hidden vehicle state at time t, u_t the control input (for example, steering and acceleration commands), and z_t the sensor measurement. The complete history available at time t is commonly written as $z_{1:t}$ and $u_{1:t}$. The filter maintains a belief distribution:

$$bel(x_t) = p(x_t \mid z_{1:t}, u_{1:t-1})$$

The belief is the probability distribution of the current state given all measurements and controls up to the present. It is not necessarily a single point estimate. A broad distribution represents high uncertainty; a narrow distribution represents greater confidence. A point estimate, such as the mean or most probable state, can be extracted when a downstream module requires a single value.

The Bayes filter is recursive: it does not need to store and reprocess the entire measurement history. It uses the previous belief, the latest control input, and the latest measurement to calculate the new belief. Its two stages are prediction (time update) and correction (measurement update).

B. Prediction: prior belief and motion model

First, the filter predicts the state forward using the motion model. The transition probability $p(x_t \mid x_{t-1}, u_{t-1})$ describes the probability of reaching state x_t from the previous state x_{t-1} under the applied control. It includes uncertainty due to imperfect dynamics, unmodelled disturbances, wheel slip, and actuator variation.

$$bel(x_t) = \int p(x_t \mid x_{t-1}, u_{t-1})\, bel(x_{t-1})\, dx_{t-1}$$

The predicted belief, also called the prior at time t, is the state distribution before incorporating the new measurement z_t. The integral marginalises over every possible previous state: each previous state contributes according to its probability and the transition model. In a discrete state space, the integral is replaced by a summation. The prediction generally becomes less certain because process noise accumulates during motion.

For a simple one-dimensional vehicle model, position may be predicted as $x_t = x_{t-1} + v_{t-1}\Delta t$, where v is velocity and Δt is the sampling interval. In practice, a car model may include acceleration, yaw rate, steering angle, and coordinate transformations. Process noise represents the mismatch between this simplified model and real vehicle behaviour.

C. Correction: measurement likelihood

When a sensor measurement arrives, the filter evaluates how likely that measurement would be for each candidate state. The measurement likelihood $p(z_t \mid x_t)$ is defined by the sensor observation model. For example, a GNSS position observation should be likely when the candidate state is near the reported position, subject to the receiver's measurement uncertainty.

$$bel(x_t) = \eta\, p(z_t \mid x_t) \,\backslash u0302 bel(x_t)$$

The product of the likelihood and predicted belief gives an unnormalised posterior. States that were predicted to be plausible and that explain the measurement well receive greater probability. The result must then be normalised so that the total probability is one.

D. Normaliser and meaning of each term

$$\eta = 1 / p(z_t) = 1 / \int p(z_t \mid x_t) \,\backslash u0302 bel(x_t)\, dx_t$$

The normaliser η is the reciprocal of the total probability (or probability density) of the observed measurement under the predicted belief. It ensures that the posterior integrates to one. The denominator is also called the evidence or marginal likelihood. It is

obtained by considering all possible states that could have generated the observation.

In summary: (i) the prior belief is what the filter believed before the latest sensor reading; (ii) the motion model propagates that belief through time; (iii) the predicted belief is the prior after motion but before sensing; (iv) the measurement likelihood scores how compatible a candidate state is with the new reading; (v) the posterior belief is the updated state distribution; and (vi) the normaliser makes the posterior a valid probability distribution. This is the general Bayes-filter cycle described in [1].

III. WHEN THE BAYES FILTER BECOMES A KALMAN FILTER

The ordinary Kalman filter is obtained when the state transition and measurement models are linear, the process and measurement noises are additive, zero-mean Gaussian and mutually independent, and the initial state belief is Gaussian. A standard discrete-time model is:

$$x_t = F_t x_{t-1} + B_t u_{t-1} + w_{t-1}$$

$$z_t = H_t x_t + v_t$$

Here F_t is the state-transition matrix; B_t maps the control input into the state; H_t maps the state into measurement space; w_t is process noise with covariance Q_t; and v_t is measurement noise with covariance R_t. The initial state is $x_0 \sim N(\mu_0, P_0)$. The Gaussian noises are $w_t \sim N(0,Q_t)$ and $v_t \sim N(0,R_t)$.

These assumptions make the mathematics tractable because a Gaussian distribution is completely specified by its mean and covariance. A linear transformation of a Gaussian variable remains Gaussian, and the product of compatible Gaussian distributions is also Gaussian. Consequently, the predicted and posterior beliefs remain Gaussian at every time step. The filter only needs to propagate a mean vector and covariance matrix rather than an arbitrary probability distribution. The integrals in the general Bayes filter can therefore be evaluated analytically, yielding the Kalman equations [2], [3].

If the dynamics or observation function is nonlinear, a Gaussian belief may no longer remain Gaussian after prediction or correction. If noise is strongly non-Gaussian, heavy-tailed, multimodal, or contains outliers, a mean and covariance may not adequately describe uncertainty. For example, a vehicle in a GNSS-denied area may have several plausible map-matched locations, producing a multimodal belief. A conventional Kalman filter can become biased or overconfident in such situations. Nonlinear approximations such as the Extended Kalman Filter (EKF) and Unscented Kalman Filter (UKF), or non-Gaussian methods such as particle filters, may be more appropriate. The EKF linearises nonlinear functions locally; the UKF propagates a carefully selected set of sigma points; particle filters represent a distribution using weighted samples.

IV. THE KALMAN FILTER

A. Prediction (time update)

The Kalman filter maintains two quantities: the estimated state mean $\hat{x}$ and its error covariance P. The prediction stage uses the motion model and control input to predict the next state:

$$\hat{x}_t^- = F_t \hat{x}_{t-1}^+ + B_t u_{t-1}$$

The superscript minus (−) denotes a prediction before the current measurement update; plus (+) denotes the estimate after correction. The equation advances the previous corrected estimate through the system dynamics. It gives the expected state, not a sensor reading.

$$P_t^- = F_t P_{t-1}^+ F_t^T + Q_t$$

This equation propagates uncertainty. F_t transforms the previous estimation error into the new state coordinates, and Q_t adds uncertainty introduced by the motion model. P is the covariance matrix: its diagonal elements are variances of individual state components, while off-diagonal elements represent correlated errors. If the vehicle moves for a longer interval or the motion model is uncertain, the predicted covariance typically increases.

B. Measurement update (correction)

The measurement update compares the actual sensor reading with the measurement predicted from the state estimate. The innovation (also called residual) is:

$$y_t = z_t - H_t \hat{x}_t^-$$

The innovation is the difference between the observed measurement and what the predicted state says the sensor should have observed. Its covariance is:

$$S_t = H_t P_t^- H_t^T + R_t$$

S_t combines predicted state uncertainty mapped into measurement space with sensor measurement-noise covariance R_t. It quantifies the expected uncertainty of the innovation. The Kalman gain is then calculated as:

$$K_t = P_t^- H_t^T S_t^{-1}$$

The Kalman gain determines how strongly the innovation changes the prediction. It balances confidence in the predicted state against confidence in the measurement. A high gain in a particular state direction means the measurement is trusted relatively strongly; a low gain means the prediction is trusted more strongly. The corrected state estimate is:

$$\hat{x}_t^+ = \hat{x}_t^- + K_t y_t$$

The estimate is moved from the prediction towards the measurement-consistent state by a gain-weighted residual. The covariance is reduced after incorporating information from the measurement. A commonly used covariance update is:

$$P_t^+ = (I - K_t H_t) P_t^-$$

For improved numerical stability, implementations often use the Joseph covariance form:

$$P_t^+ = (I - K_t H_t) P_t^- (I - K_t H_t)^T + K_t R_t K_t^T$$

The Joseph form better preserves symmetry and positive semi-definiteness in finite-precision computation. I is the identity matrix and the superscript T denotes matrix transpose. These prediction and correction equations form the standard linear Kalman filter [2], [3].

C. Interpreting the Kalman gain

The gain is not a fixed percentage; it is calculated at each update from P_t^-, H_t, and R_t. If the sensor is very noisy, R_t is large. This increases S_t and generally reduces K_t, so the filter makes only a small correction and relies more on the motion prediction. If the sensor is very accurate, R_t is small. The gain generally increases, so the estimate moves more strongly towards the measurement.

For a scalar example with a direct position measurement (H = 1), $K = P^-/(P^- + R)$. If predicted variance $P^- = 4$ and measurement variance R = 1, K = 0.8: the measurement receives substantial weight. If R = 16, K = 0.2: the filter relies more heavily on its prediction. If prediction uncertainty is high while measurement uncertainty is low, the gain also increases. Thus, the gain responds to relative uncertainty, not sensor accuracy alone.

V. RELEVANCE TO AN AUTONOMOUS CAR

A. Where filtering fits in the driving pipeline

A typical autonomous-driving pipeline contains sensor acquisition, calibration and synchronisation, perception, localisation and mapping, prediction, planning, and vehicle control. State estimation sits across the perception and localisation stages and supplies a temporally consistent estimate to downstream modules. It can operate as an early sensor-fusion block that combines measurements from different sensors, or as part of a tightly coupled estimator that uses raw or intermediate sensor information.

A practical vehicle-state vector might be $x = [X, Y, v, \psi, r]^T$, where X and Y are position in a chosen reference frame, v is speed, ψ is heading, and r is yaw rate. A more complete estimator may include longitudinal and lateral velocity, accelerometer and gyroscope biases, and steering parameters. The precise state and model depend on vehicle dynamics, sensor placement, update rates, and the intended operating domain.

B. Example: IMU, wheel odometry and GNSS

The IMU provides high-rate angular velocity and acceleration measurements. These are useful for short-term motion propagation, but small biases accumulate through integration and cause drift. Wheel-speed sensors provide vehicle-speed information, but their readings can be wrong during wheel slip, braking, or uneven road contact. GNSS provides an absolute or globally referenced position, but its update rate is often lower and its accuracy can degrade near buildings, trees, tunnels, or under multipath conditions.

A filter can propagate the vehicle state at the IMU rate using the motion model, then correct the prediction when wheel-speed, GNSS, or other observations arrive. In a simplified linear example, the state prediction uses F and Q; a GNSS position update uses H to select the position components and R to represent GNSS uncertainty. A wheel-speed update can constrain the velocity component. Sensor-specific measurement updates can be applied sequentially, provided their timing and noise models are handled correctly. The filter should increase R or reject measurements when quality indicators show that a sensor is unreliable.

For a realistic car, the motion and observation equations are generally nonlinear. For example, planar vehicle motion includes terms such as $\dot{X} = v \cos \psi$ and $\dot{Y} = v \sin \psi$. An EKF may linearise these equations around the current estimate, while an UKF can approximate the transformed distribution without explicitly calculating Jacobians. A production system may also use factor-graph optimisation or other estimators. The basic Bayes-filter principle remains the same: predict from dynamics, compare with observations, and update uncertainty.

C. Interface with downstream modules

The filter output is a timestamped state estimate and uncertainty, not a driving decision. Perception can use the ego-motion estimate to compensate for vehicle movement between camera or LiDAR frames and to transform detected objects into a common coordinate frame. Localisation uses the estimate to track the vehicle relative to a map. Object tracking uses ego-motion to distinguish an object's movement from apparent movement caused by the car itself. Prediction estimates how other road users may move; planning selects a safe trajectory; and control tracks that trajectory.

A poor state estimate can therefore propagate errors through the entire pipeline. A heading error can rotate the perceived road geometry; a velocity error can distort time-to-collision calculations; and position drift can misalign the vehicle with lane boundaries or map features. The covariance is valuable because downstream modules can account for confidence rather than treating every estimate as exact. Filtering does not eliminate sensor failure: robust outlier detection, sensor-health monitoring, calibration, time synchronisation, and safe fallback behaviour remain necessary.

VI. LIMITATIONS AND PRACTICAL CONSIDERATIONS

The classical Kalman filter is optimal in the minimum-mean-square-error sense under its linear-Gaussian assumptions. It is not universally optimal for arbitrary vehicle systems. Model mismatch, incorrect noise covariances, unmodelled delays, sensor bias, asynchronous measurements, and outliers can degrade performance. Underestimating Q or R can make the filter overconfident; overestimating them can make it unnecessarily sluggish. Covariance tuning should be based on sensor specifications, experimental data, and validation across operating conditions.

The estimator must also handle coordinate frames, units, timestamps, and data association consistently. A GNSS reading in a global geodetic frame cannot be combined directly with a local vehicle-frame state without a valid transformation. Measurements should be associated with the correct acquisition time; delayed measurements may require buffering and re-propagation. Heading is an angular variable and must be wrapped appropriately to avoid artificial discontinuities around $\pm\pi$. These engineering details are essential for reliable deployment.

VII. CONCLUSION

The Bayes filter provides the general framework for estimating a hidden vehicle state from uncertain motion and sensor information. Its prediction step uses the motion model to form a prior belief, while its correction step weights that belief by the measurement likelihood and normalises it to obtain a posterior. Under linear dynamics, linear measurements, Gaussian noise, and a Gaussian initial belief, the distribution remains Gaussian and the recursive update has the closed-form Kalman filter solution. The Kalman gain controls the balance between prediction and measurement according to their relative uncertainty. In an autonomous car, filtering supports localisation, ego-motion estimation, sensor fusion, object tracking, and the state information required by planning and control. The classical Kalman filter is therefore a foundational estimator, while nonlinear or non-Gaussian driving conditions motivate extensions such as EKF, UKF, and particle filtering.

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